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Apparatus for providing parallel filtering

Abstract

In at least one embodiment, a system comprising a first circuit and a second circuit is provided. The first circuit includes a first filter filtering a first input signal and generating a first filtered output signal. The first circuit includes a first comparator comparing the first filtered output signal to a first threshold value and providing a first output signal indicative of a failure condition to one or more first processors. The second circuit includes a second filter filtering the first input signal and generating a second filtered output signal to filter the first input signal in less time than the first filter filtering the first input signal. The second circuit includes a second comparator comparing the second filtered output signal to a second threshold value and providing a second output signal indicative of the failure condition to one or more second processors.

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Background/Summary

TECHNICAL FIELD

(1) Aspects disclosed herein generally relate to a system for providing parallel filtering. These aspects and others will be discussed in more detail herein.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 depicts an apparatus for providing parallel filtering in accordance with one embodiment; (2) FIG. 2 depicts an equivalent circuit that indicates the manner in which a first time constant and a second time constant are determined in accordance with one embodiment; and (3) FIG. 3 depicts various measurements of a sensed voltage and reaction times for disabling a charging operation.

DETAILED DESCRIPTION

(4) As required, detailed embodiments of the present invention are disclosed herein; however, it is to be understood that the disclosed embodiments are merely exemplary of the invention that may be embodied in various and alternative forms. The figures are not necessarily to scale; some features may be exaggerated or minimized to show details of particular components. Therefore, specific structural and functional details disclosed herein are not to be interpreted as limiting, but merely as a representative basis for teaching one skilled in the art to variously employ the present invention.

(5) Reference will now be made in detail to embodiments, examples of which are illustrated in the accompanying drawings. In the following detailed description, numerous specific details are set

forth in order to provide a thorough understanding of the various described embodiments.

However, it will be apparent to one of ordinary skill in the art that the various described embodiments may be practiced without these specific details. In other instances, well-known methods, procedures, components, circuits, and networks have not been described in detail so as not to unnecessarily obscure aspects of the embodiments.

(6) It is to be understood that the disclosed embodiments are merely exemplary and that various and alternative forms are possible. The figures are not necessarily to scale; some features may be exaggerated or minimized to show details of particular components. Therefore, specific structural and functional details disclosed herein are not to be interpreted as limiting, but merely as a representative basis for teaching one skilled in the art to variously employ embodiments according to the disclosure.

(7) “One or more” and/or “at least one” includes a function being performed by one element, a function being performed by more than one element, e.g., in a distributed fashion, several functions being performed by one element, several functions being performed by several elements, or any combination of the above.

(8) It will also be understood that, although the terms first, second, etc. are, in some instances, used herein to describe various elements, these elements should not be limited by these terms. These terms are only used to distinguish one element from another. For example, a first contact could be termed a second contact, and, similarly, a second contact could be termed a first contact, without departing from the scope of the various described embodiments. The first contact and the second contact are both contacts, but they are not the same contact.

(9) The terminology used in the description of the various described embodiments herein is for the purpose of describing particular embodiments only and is not intended to be limiting. As used in the description of the various described embodiments and the appended claims, the singular forms “a”, “an” and “the” are intended to include the plural forms as well, unless the context clearly indicates otherwise. It will also be understood that the term “and/or” as used herein refers to and encompasses any and all possible combinations of one or more of the associated listed items. It will be further understood that the terms “includes,” “including,” “comprises,” and/or “comprising,” when used in this specification, specify the presence of stated features, integers, steps, operations, elements, and/or components, but do not preclude the presence or addition of one or more other features, integers, steps, operations, elements, components, and/or groups thereof.

(10) As used herein, the term “if” is, optionally, construed to mean “when” or “upon” or “in response to determining” or “in response to detecting,” depending on the context. Similarly, the phrase “if it is determined” or “if [a stated condition or event] is detected” is, optionally, construed to mean “upon determining” or “in response to determining” or “upon detecting [the stated condition or event]” or “in response to detecting [the stated condition or event],” depending on the context.

(11) Aspects disclosed herein generally provide a system that performs parallel filtering for a sensing circuit. The sensing circuit senses an incoming voltage for example an on-board charging device (OBC). The disclosed system comprises a first circuit comprising at least a first filter and a second circuit comprising at least a second filter. Each of the first circuit and the second circuit receive a first input signal that is indicative of at least a sensed voltage condition in the system. The first filter operates a first operating frequency, and the second filter operates at a second operating frequency. The second operating frequency of the second filter is greater than the first operation frequency thereby enabling the second circuit to detect a failure condition of the system faster than the first circuit detecting the failure condition. Generally, the second circuit outputs an output signal to transmit to one or more processors (or microprocessors and/or digital signal processors (DSPs)) in the system to quickly disable the system to mitigate the failure condition.

(12) The first circuit also detects the failure condition however such detection is performed at a time that is greater than that of the second circuit detecting the failure condition due to the first

operating frequency being less than the second operating frequency. In this case, the first circuit detects the failure condition to report out the detected failure condition to other processors (e. g., in the vehicle for diagnostic purposes and/or to be used as a redundant check on the first circuit). Thus, it may not be necessary for the first circuit to detect the failure condition at the same rate as that of the second circuit.

(13) The first operating frequency for the first filter and the second operating frequency for the second filter are based on a first time constant and a second time constant, respectively. As disclosed herein, a correlation exists between the electronics used to form the first circuit and the second circuit and the various time constants at which the first and second filters operate. For example, the first time constant may be derived based on values for a first resistive and capacitive network of the first filter. Similarly, the second time constant may be derived based on values for a second resistive and capacitive network of the second filter. Other implementations that are used to detect failure conditions via parallel filtering methods in a system utilize buffer circuits (or operational amplifiers) to isolate the filters from one another. The disclosed system eliminates the need for the buffer circuits. In addition, the disclosed system provides a failure detection mechanism that may be applied to any number of applications that have differing delay times, alarm thresholds, responses to ramp or step behaviors, etc. In addition, the disclosed system may be well suited for applications that require redundant detection mechanisms in place to provide any number of parallel measurements.

(14) FIG. 1 depicts a system **100** (or apparatus) for providing parallel filtering and for detecting failure conditions in accordance with one embodiment. The system **100** generally includes a first circuit **102**, a second circuit **104**. It is recognized that the system **100** may include any number of circuits that operate similar to any one of the first circuit **102** and/or the second circuit **104**. In general, each of the first circuit **102** and the second circuit **104** are configured to detect fault conditions (or failure condition) (e.g., over voltage condition, undervoltage condition, over current condition and/or under current condition) attributed to a sensed input (e.g., sensed voltage or sensed current). In one example, the system **100** detects fault conditions for an on-board charger (OBC) **103** in a vehicle **105**. It is contemplated that the system **100** may be used for any application that is configured to detect a fault condition.

(15) The system **100** includes a sense input circuit **130** that provides a first input signal indicative of the sensed voltage to the first circuit **102** and the second circuit **104**. The first circuit **102** includes a first filter **110** and a first comparator **114**. A first resistive and capacitive network **113** generally forms the first filter **110**. The first resistive and capacitive network **113** includes resistor **R1** and capacitor **C1**. In one example, the first filter **110** is a low pass filter. It is recognized that the first filter **110** may be a high pass filter, band pass filter, etc. The first filter **110** filters the incoming sensed voltage and provides an output to the first comparator **114**. The first comparator **114** compares the output from the first filter **110** to a threshold which then outputs a first output signal indicative of a detected fault based on the comparison.

(16) The first filter **110** operates at a first frequency when performing the filtering operation. The relevance of the first frequency will be more discussed in more detail below. The first circuit **102** provides an output (or the first output signal) to one or more first processors **120** (“the first processor **120**”) indicating the presence of a fault detection condition in the system **100**. The first processor **120** may include one or more first microcontrollers or one or more first digital signal processors (DSPs). It is recognized that the first DSPs may operate faster than the first microcontrollers. The first processor **120** reports the presence of a failure as a diagnostic feature and may transmit an indication of the detected fault condition to alert users of the condition. The fault condition may correspond to an over voltage condition, an under-voltage condition, an over current condition, or an under current condition. In the case of a detected over current condition or an under current condition, the first circuit **102** reports such a condition to the first DSPs since the first DSPs respond faster than the first microcontrollers. In the case of a detected over voltage

condition or under voltage condition, the first circuit **102** reports such a condition to the first microcontrollers as the faults related to over and under voltage conditions may not have the same priority as that of the detected over and under current conditions.

(17) In a similar fashion, the second circuit **104** also provides an output (or the second output signal) to one or more second processors **122** (i.e., “the second processor **122**”) in response to detecting the failure condition. The second processor **122** may include one or more second microcontrollers or one or more second digital signal processors (DSPs). It is recognized the second DSPs may operate faster than that of the second microcontrollers. For example, the second circuit **104** receives the first input signal from the sense input circuit **130** that is indicative of the sensed voltage (or current) in the system **100**. The second circuit **104** includes a second filter **112a** and a second comparator **116a**. A second resistive and capacitive network **115** generally forms the second filter **112a**. The second resistive and capacitive network **115** includes resistor **R2a** and capacitor **C2a**. In one example, the second filter **112a** is a low pass filter. It is recognized that the second filter **112a** may be a high pass filter, band pass filter, etc. The second filter **112a** filters the incoming sensed voltage and provides an output (e.g., a second output signal) to the second comparator **116a**. The second comparator **116a** compares the output from the second filter **112a** to a threshold which then outputs the second output signal indicative of a detected fault based on the comparison.

(18) The second filter **112a** operates at a second frequency when performing the filtering operation. The second frequency may be greater than the first frequency of that used by the first filter **110**. The second circuit **104** provides the second output signal to one or more second processors **122** indicating the presence of the fault detection condition in the system **100**. The second processor **122** disables the system **100** to prevent the fault condition from damaging electronics or other devices in the system **100** in response to detecting the fault condition. In one example, the second processor **122** (e.g., the second DSP) may disable the charging operation being performed between the OBC **103** and the vehicle **105** in matter of microseconds. The fault condition may correspond to an over voltage condition, an under-voltage condition, an over current condition, or an under current condition.

(19) Generally, the second circuit **104** is tasked with detecting the fault condition as quickly as possible since the second circuit **104** transmits the second output signal to the second processor **122** to disable the system **100** (or OBC **103**) when a fault condition is detected. For example, the second processor **122** may control any number of direct current (DC) to DC converters (not shown) that converts a high voltage incoming signal during a charging operation of the vehicle **105**. Upon detecting or receiving the second output signal from the second circuit **104**, the second processor **122** disables the battery charging operation given the high voltage conversion that takes place to mitigate any potential issues. The first processor **120** as noted above in connection with the first circuit **102** may detect the presence of the failure condition in response to receiving the first output signal from the first circuit **102** and notify other microprocessors (or controllers) in the vehicle **105** that a failure condition (e.g., over voltage condition or undervoltage condition) has been detected in connection with the vehicle charging operation. However, the first processor **120** may not be directly responsible for controlling the various DC/DC converters that perform the energy conversion during the battery charging operations and may simply log the presence of the fault detection in the event this condition is present so that other measures may be taken. If however, the first circuit **102** is arranged to detect an over current or under current condition, then the first circuit **102** reports such a condition to both the first microcontroller and the first DSP of the first processor **120**. The first output signal as output by the first circuit **102** may serve as a redundant output and be utilized to disable the charging operation in the event the second processor **122** has malfunctioned.

(20) Thus, by operating the second filter **112a** of the second circuit **104** at a second frequency that is greater than that of the first frequency for the first circuit **102**, the second circuit **104** is configured to process (e.g., filter the first input signal indicative of the sensed voltage or current

with the second filter **112a**) faster than that of the first circuit **102** (e.g., faster than the first filter **110**) and hence detect the fault condition earlier in time when viewed in reference to the first circuit **102**. The second circuit **104** is also able to report out the fault detection the second processor **122** faster than the first circuit **102** is able to report out the fault condition to the first processor **120**.

(21) The second circuit **104** includes a third filter **112b** and a third comparator **116b**. Each of the first filter **110**, the second filter **112a**, and the third filter **112b** are included in the system **100** since each input these filters **110**, **112a**, and **112b** provide are independent readings from one another and therefore provide redundant measuring circuits. A third resistive and capacitive network **117** generally forms the third filter **112b**. The third resistive and capacitive network **117** includes resistor **R2b** and capacitor **C2b**. In one example, **R2b** is equal to **R2a** and **C2b** is equal to **C2a**. In another example, the third filter **112c** is a low pass filter. It is recognized that the third filter **112c** may be a high pass filter, band pass filter, etc. The third filter **112b** filters the incoming sensed voltage and provides an output to the third comparator **116b**. The third comparator **116b** compares the output from the third filter **112a** to a threshold which then outputs a third output signal indicative of a detected fault based on the comparison.

(22) The third filter **112b** operates at a third frequency when performing the filtering operation. The third frequency may be greater than the first frequency of that used by the first filter **110**. In one example, the third frequency may be similar (or equal) to the second frequency of the second filter **112a**. The third circuit **112b** also provides the third output signal to the second processor **122** indicating the presence of the fault detection condition in the system **100**. As noted above, the second processor **122** disables the system **100** to prevent the fault condition from damaging electronics or other devices in the system **100** in response to detecting the fault condition.

(23) The second circuit **104** is tasked with detecting the fault condition as quickly as possible since the second circuit **104** transmits the second output signal to the second processor **122** to disable the system **100** (or OBC **103**) when a fault condition is detected. Thus, by operating the third filter **112b** of the second circuit **104** at the second frequency that is greater than that of the first frequency for the first circuit **102**, the second circuit **104** is configured to process (e.g., filter the first input signal indicative of the sensed voltage with the second filter **112a**) faster (or in less than time) than that of the first circuit **102** and hence detect the fault condition earlier in time when viewed in reference to the first circuit **102**. The second circuit **104** is also able to report out the fault detection the second processor **122** faster than the first circuit **102** is able to report out the fault condition to the first processor **120**. In one example, the second processor **122** disables the OBC **103** based on receiving the second output signal from the second circuit **104** prior to the first processor **120** reporting the failure condition based on receiving the first output from the first circuit **102** given that the second filter **112a** and the third filter **112b** operate at the second frequency (and also have a larger frequency range).

(24) Input threshold resistors **Rin1**, **Rin2**, and **Rin3** for the first comparator **114**, the second comparator **116a**, and the third comparator **116b**, respectively, provide a threshold voltage, **V.sub.TH** (or threshold current) for each of the first comparator **114**, the second comparator **116a**, and the third comparator **116b**. Thus, the resistance values for the input threshold resistors **Rin1**, **Rin2**, and **Rin2** may be selected to enable the first comparator **114**, the second comparator **116a**, and the third comparator **116b** to provide an output that is indicative of an overvoltage condition, an undervoltage condition, the overcurrent condition, and/or the undercurrent condition. While two circuits **102** and **104** are shown, it is recognized that any number of circuits may be included in the system **100** to provide a plurality of outputs that correspond to a detected overvoltage condition.

(25) In general, the operating frequencies (e.g., the first frequency for the first filter **110**, the second frequency for the second filter **112a**, and the third frequency for the third filter **112b**) can be calculated by the following equation which is a function of the time constant (e.g., "**τ**"):

(26) $f_c = \frac{1}{2\pi RC} = \frac{1}{2\pi \tau}$ (1)

(27) Thus, in this regard, the selection of particular resistance and capacitance values (e.g., R1 and C1) for the first resistive capacitive network **113** at least partially establish the first frequency (or f.sub.1). Similarly, the particular resistance and capacitance values (e.g., R2a, C2a and R2b, C2b) for the second and third capacitive networks **115** and **117**, respectively, at least partially form the second frequency for the second filter **112a** and the third filter **112b**. In one example, the second filter **112a** and the third filter **112b** may both operate at the second frequency. In this regard, the resistance value for R2a and the capacitance value for C2a of the second resistive and capacitive network **115** are equal to the resistance value R2b and the capacitance value for C2b of the third resistive and capacitive network **117**. In another example, it is recognized that the second filter **112a** and the third filter **112b** may operate at different frequencies from one another but may still be arranged to operate faster than that of the first filter **113**. In this regard, the resistance values for R2a and R2b may be different from one another and/or the capacitance values for C2a and C2b may be different from one another. By selecting the various values for the RC networks for the first filter **110**, the second filter **112a**, and/or the third filter **112**, to achieve the desired frequencies to control the speed at which the first circuit **102** and the second circuit **104** measure the sensed voltage, such an implementation yields increased performance no longer requires additional electronics and achieves a cost saving.

(28) FIG. 2 depicts an equivalent circuit **200** that indicates the manner in which a first time constant (e.g. t1) and a second time constant (e.g. t2) are determined in accordance with one embodiment. The first frequency for the first filter **110** is based on the first time constant (e.g., t1). Similarly, the second frequency for the second filter **112a** and the third filter **112b** is based on the second time constant (e.g. t2) assuming that both the second filter **112a** and the third filter **112b** operate at the same frequency. As noted above, the second filter **112a** and the third filter **112b** may operate at different frequencies from one another so long as such frequencies enable the second filter **112a** and the third filter **112b** to operate (or perform the filtering operation) faster than that of the first filter **110**.

(29) The circuit **200** generally corresponds to aspects shown in connection with the first circuit **102**, the second circuit **104**, and the sense input circuit **130**. For example, the circuit **200** includes the resistors R1, R2, Rd1 and Rd2 and capacitors C1, C2. The first time constant, t1 is generally defined (or formed) based on the following equation:

$$\tau_{\text{sub.1}} = C1 * (R1 + R_{\text{sub.th}}) \quad (2)$$

(30) where $R_{\text{sub.th}} = Rd1 \parallel Rd2$.

(31) The second time constant, $\tau_{\text{sub.2}}$ is generally defined (or formed) based on the following equation:

$$\tau_{\text{sub.2}} = C2 * (R2 + R_{\text{sub.th}} \parallel R1) \quad (3)$$

(32) where $C2 = 2 * C2a$ and $R2 = R2a / 2$.

(33) Conversely, the first time constant $\tau_{\text{sub.1}}$ and the second time constant $\tau_{\text{sub.2}}$ may be provided as follows: $\tau_{\text{sub.1}} = C1 * (R1)$ or $\tau_{\text{sub.2}} = C2a * (R2a)$ (or $\tau_{\text{sub.2}} = C2b * (R2b)$). Thus, as shown, the various resistance values for the resistors R1, R2a, Rd1, Rd2 and the capacitance values for the capacitors C1 and C2a may be used to generate the first time constant and the second time constant based on the equations shown above. The first time constant and the second time constant may be used based on equation (1) above to derive the first frequency and the second frequency for the first filter **110** and the second filter **112a** and consequently the third filter **112b**. Given the foregoing, it can be seen that the first time constant is greater than the second time constant (or $\tau_{\text{sub.1}} > \tau_{\text{sub.2}}$) in order for the first frequency for the first filter **110** to be less than the second frequency for the second filter **112a** and the third filter **112b** (e.g., in order for the second filter **112a** (or the third filter **112b**) to operate faster than the first filter **110**).

(34) FIG. 3 depicts a table **200** illustrating various measurements of the sensed voltage and reaction times for disabling a charging operation. In general, the table **200** includes first column **202**, second column **203**, third column **204**, and fourth column **206**. The first column **202** generally corresponds

to various sensed signals for voltages or currents. The second column **203** generally corresponds to various signal parameters. The third column **204** generally corresponds to first measurements as performed by a system that does not include aspects of the disclosed system **100**. The fourth column **206** generally corresponds to second measurements as performed by the disclosed system **100**.

(35) In one example, the sensed signals as shown in first column **202** of the table **200** correspond to sensed voltages or currents on a HV network in the vehicle **105**. In another example, the sensed signals in the first column **202** of the table **200** correspond to sensed voltages or currents on a low voltage (LV) network in the vehicle **105**. The second column **203** depicts various parameters that correspond to the sensed characteristics (e.g., voltage or current). The third column **204** depicts various measurement times as performed by a different system (not the disclosed system **100**) for each of the sensed voltages and currents that are shown in the first column **202**. The fourth column **206** depicts various measurement times for each of the sensed voltages and currents that are shown in the first column **202** as performed by the first circuit **102** of the system **100**.

(36) As generally shown at **250**, the different system measures the sensed voltage on the HV network at 51 μ s whereas the system **100** (or the second circuit **104**) as disclosed herein generally measures the sensed voltage on the HV network at 12 μ s. As generally shown at **252**, the different system measures/detects the sensed current on the HV network at 356 μ s whereas the system **100** (or the second circuit **104**) as disclosed herein measures/detects the sensed current on the HV network at 33 μ s. As generally shown at **254**, the different system generally measures the sensed voltage on the LV network at 736 μ s whereas the system **100** (or the second circuit **104**) as disclosed herein generally measures/detects the sensed voltage on the LV network at 28 μ s. As generally shown at **256**, the different system generally measures/detects the sensed current on the HV network at 202 μ s whereas the system **100** (or the second circuit **104**) as disclosed herein measures/detects the sensed current on the HV network at 40 μ s. Thus, the system **100** exhibits a faster reaction period over the different system in any one or more cases.

(37) Item 1. A system comprising a first circuit and a second circuit. The first circuit comprising a first filter filtering a first input signal indicative of at least a sensed voltage condition in the system, the first filter generating a first filtered output signal in response to filtering the first input signal at a first frequency. The first circuit comprising a first comparator comparing the first filtered output signal to a first threshold value and providing a first output signal indicative of a failure condition to one or more first processors to report the failure condition based on the comparison of the first filtered output signal to the first threshold value. The second circuit comprising a second filter filtering the first input signal indicative of the at least the sensed voltage condition within the system, the second filter generating a second filtered output signal in response to filtering the first input signal at a second frequency that is greater than the first frequency enabling the second filter to filter the first input signal in less time than the first filter filtering the first input signal. The second circuit comprising a second comparator comparing the second filtered output signal to a second threshold value and providing a second output signal indicative of the failure condition to one or more second processors disabling the system prior to the one or more first processors reporting the failure condition based on the comparison of the second filtered output signal to the second threshold value.

(38) Item 2. According to item 1, the second circuit comprising a third filter filtering the first input signal indicative of the at least the sensed voltage condition, and wherein the third filter generating a third filtered output signal in response to filtering the first input signal at the second frequency that is greater than the first frequency enabling the second filter to filter the first input signal in less time than the first filter filtering the first input signal.

(39) Item 3. According to item 2, the second circuit comprising a third comparator comparing the third filtered output signal to a third threshold value and providing a third output signal indicative of the failure condition to the one or more second processors, and the one or more second

processors disabling the system prior to the one or more first processors reporting the failure condition based on the comparison of the second filtered output signal to the third threshold voltage.

- (40) Item 4. According to item 3, the third threshold value for the third comparator is different than the second threshold value for the second comparator.
- (41) Item 5. According to item 4, the second comparator outputs a signal indicative of an overvoltage condition based on the comparison the second filtered output signal to the second threshold value.
- (42) Item 6. According to item 4, the third comparator outputs a signal indicative of an undervoltage condition based on the comparison of the third filtered output signal to the third threshold value.
- (43) Item 7. According to item 1, the first frequency for the first filter is based on a first time constant and the second frequency for the second filter is based on a second time constant, and wherein the first time constant is different from the second time constant.
- (44) Item 8. According to item 7, the first time constant is greater than the second time constant.
- (45) Item 9. According to item 7, further comprising a sense input circuit comprising at least one sensing resistor being electrically connected with the first filter and the second filter providing the first input signal that is indicative of the sensed voltage condition.
- (46) Item 10. According to item 9, the first time constant and the second time constant are each partially based on a resistance value of the least one sensing resistor.
- (47) Item 11. According to item 7, the first filter is formed from a first resistive and capacitive network and the second filter is formed from a second resistive and capacitive network.
- (48) Item 12. According to item 11, the first time constant is at least partially based on a first resistance value and a first capacitance value of the first resistive and capacitive network, and the second time constant is at least partially based on a second resistance value and a second capacitance value of the second resistive and capacitive network.
- (49) Item 13. According to item 1, the first filter is a low pass filter.
- (50) Item 14. According to item 1, wherein the second filter is a low pass filter.
- (51) Item 15. A system comprising a first circuit and a second circuit. The first circuit comprising a first filter filtering a first input signal indicative of at least a sensed voltage condition, the first filter generating a first filtered output signal in response to filtering the first input signal at a first frequency, and a first comparator comparing the first filtered output signal to a first threshold value and providing a first output signal indicative of a failure condition to one or more first processors to report the failure condition based on the comparison of the first filtered output signal to the first threshold value. The second circuit comprising a second filter filtering the first input signal indicative of the at least the sensed voltage condition within the system, the second filter generating a second filtered output signal in response to filtering the first input signal at a second frequency that is greater than the first frequency enabling the second filter to filter the first input signal in less time than the first filter filtering the first input signal. The second circuit comprising a second comparator comparing the second filtered output signal to a second threshold value and providing a second output signal indicative of the failure condition to one or more second processors. The first frequency of the first filter is based at least on a first time constant and the second frequency of the second filter is based at least on a second time constant, and the second time constant is less than the first time constant causing the second filter to filter the first input signal in less time than the first filter filtering the first input signal.
- (52) Item 16. According to item 15, the system comprising a sense input circuit comprising at least one sensing resistor being electrically connected with the first filter and the second filter providing the first input signal that is indicative of the sensed voltage condition.
- (53) Item 17. According to item 16, the first time constant and the second time constant are each partially based on a resistance value of the least one sensing resistor.

(54) Item 18. According to item 15, the first filter is formed from a first resistive and capacitive network, and the second filter is formed from a second resistive and capacitive network.

(55) Item 19. According to item 18, the first time constant is at least partially based on a first resistance value and a first capacitance value of the first resistive and capacitive network and the second time constant is at least partially based on a second resistance value and a second capacitance value of the second resistive and capacitive network.

(56) Item 20. A system comprising a first filter, a first comparator, and a second filter. The first filter filtering a first input signal indicative of at least a sensed voltage condition in the system, the first filter generating a first filtered output signal in response to filtering the first input signal at a first frequency. The first comparator comparing the first filtered output signal to a first threshold value and providing a first output signal indicative of a failure condition to one or more first processors to report the failure condition based on the comparison of the first filtered output signal to the first threshold value. The second filter filtering the first input signal indicative of the at least the sensed voltage condition within the system, the second filter generating a second filtered output signal in response to filtering the first input signal at a second frequency that is different than the first frequency enabling the second filter to filter the first input signal in less time than the first filter filtering the first input signal. The second comparator comparing the second filtered output signal to a second threshold value and providing a second output signal indicative of the failure condition to one or more second processors disabling the system prior to the one or more first processors reporting the failure condition based on the comparison of the second filtered output signal to the second threshold value.

(57) While exemplary embodiments are described above, it is not intended that these embodiments describe all possible forms of the invention. Rather, the words used in the specification are words of description rather than limitation, and it is understood that various changes may be made without departing from the spirit and scope of the invention. Additionally, the features of various implementing embodiments may be combined to form further embodiments of the invention.

Claims

1. A system comprising a first circuit comprising: a first filter filtering a first input signal indicative of at least a sensed voltage condition in the system, the first filter generating a first filtered output signal in response to filtering the first input signal at a first frequency, and a first comparator comparing the first filtered output signal to a first threshold value and providing a first output signal indicative of a failure condition to one or more first processors to report the failure condition based on the comparison of the first filtered output signal to the first threshold value; and a second circuit comprising: a second filter filtering the first input signal indicative of the at least the sensed voltage condition within the system, the second filter generating a second filtered output signal in response to filtering the first input signal at a second frequency that is greater than the first frequency enabling the second filter to filter the first input signal in less time than the first filter filtering the first input signal, and a second comparator comparing the second filtered output signal to a second threshold value and providing a second output signal indicative of the failure condition to one or more second processors disabling the system prior to the one or more first processors reporting the failure condition based on the comparison of the second filtered output signal to the second threshold value.

2. The system of claim 1, wherein the second circuit comprises a third filter filtering the first input signal indicative of the at least the sensed voltage condition, and wherein the third filter generating a third filtered output signal in response to filtering the first input signal at the second frequency that is greater than the first frequency enabling the second filter to filter the first input signal in less time than the first filter filtering the first input signal.

3. The system of claim 2, wherein the second circuit comprises a third comparator comparing the

third filtered output signal to a third threshold value and providing a third output signal indicative of the failure condition to the one or more second processors, and wherein the one or more second processors disabling the system prior to the one or more first processors reporting the failure condition based on the comparison of the second filtered output signal to the third threshold voltage.

4. The system of claim 3, wherein the third threshold value for the third comparator is different than the second threshold value for the second comparator.

5. The system of claim 4, wherein the second comparator outputs a signal indicative of an overvoltage condition based on the comparison of the second filtered output signal to the second threshold value.

6. The system of claim 4, wherein the third comparator outputs a signal indicative of an undervoltage condition based on the comparison of the third filtered output signal to the third threshold value.

7. The system of claim 1, wherein the first frequency for the first filter is based on a first time constant and the second frequency for the second filter is based on a second time constant, and wherein the first time constant is different from the second time constant.

8. The system of claim 7, wherein the first time constant is greater than the second time constant.

9. The system of claim 7 further comprising a sense input circuit comprising at least one sensing resistor being electrically connected with the first filter and the second filter providing the first input signal that is indicative of the sensed voltage condition.

10. The system of claim 9, wherein the first time constant and the second time constant are each partially based on a resistance value of the least one sensing resistor.

11. The system of claim 7, wherein the first filter is formed from a first resistive and capacitive network and the second filter is formed from a second resistive and capacitive network.

12. The system of claim 11, wherein the first time constant is at least partially based on a first resistance value and a first capacitance value of the first resistive and capacitive network, and wherein the second time constant is at least partially based on a second resistance value and a second capacitance value of the second resistive and capacitive network.

13. The system of claim 1, wherein the first filter is a low pass filter.

14. The system of claim 1, wherein the second filter is a low pass filter.

15. A system comprising: a first circuit comprising: a first filter filtering a first input signal indicative of at least a sensed voltage condition, the first filter generating a first filtered output signal in response to filtering the first input signal at a first frequency, and a first comparator comparing the first filtered output signal to a first threshold value and providing a first output signal indicative of a failure condition to one or more first processors to report the failure condition based on the comparison of the first filtered output signal to the first threshold value; and a second circuit comprising: a second filter filtering the first input signal indicative of the at least the sensed voltage condition within the system, the second filter generating a second filtered output signal in response to filtering the first input signal at a second frequency that is greater than the first frequency enabling the second filter to filter the first input signal in less time than the first filter filtering the first input signal, and a second comparator comparing the second filtered output signal to a second threshold value and providing a second output signal indicative of the failure condition to one or more second processors, wherein the first frequency of the first filter is based at least on a first time constant and the second frequency of the second filter is based at least on a second time constant, and wherein the second time constant is less than the first time constant causing the second filter to filter the first input signal in less time than the first filter filtering the first input signal.

16. The system of claim 15 comprising a sense input circuit comprising at least one sensing resistor being electrically connected with the first filter and the second filter providing the first input signal that is indicative of the sensed voltage condition.

17. The system of claim 16, wherein the first time constant and the second time constant are each partially based on a resistance value of the least one sensing resistor.
18. The system of claim 15, wherein the first filter is formed from a first resistive and capacitive network, and wherein the second filter is formed from a second resistive and capacitive network.
19. The system of claim 18, wherein the first time constant is at least partially based on a first resistance value and a first capacitance value of the first resistive and capacitive network and the second time constant is at least partially based on a second resistance value and a second capacitance value of the second resistive and capacitive network.
20. A system comprising: a first filter filtering a first input signal indicative of at least a sensed voltage condition in the system, the first filter generating a first filtered output signal in response to filtering the first input signal at a first frequency, and a first comparator comparing the first filtered output signal to a first threshold value and providing a first output signal indicative of a failure condition to one or more first processors to report the failure condition based on the comparison of the first filtered output signal to the first threshold value; and a second filter filtering the first input signal indicative of the at least the sensed voltage condition within the system, the second filter generating a second filtered output signal in response to filtering the first input signal at a second frequency that is different than the first frequency enabling the second filter to filter the first input signal in less time than the first filter filtering the first input signal, and a second comparator comparing the second filtered output signal to a second threshold value and providing a second output signal indicative of the failure condition to one or more second processors disabling the system prior to the one or more first processors reporting the failure condition based on the comparison of the second filtered output signal to the second threshold value.
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